ANTIBIOTICS

SUZON
CHEM-50

Antibiotics:

The term antibiotic (Anti- Against, biolife) only applies to antibacterial that are produced by one organism (Bacteria, fungi) or different living organism that either kill or inhibit the metabolism, growth of the bacteria.



E. coli

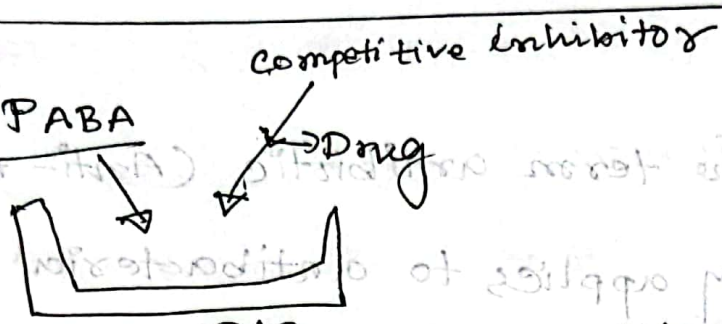
→ Staphilococcus.

① Bacteriostatic
② Bactericidial.

Competitive Inhibitor:

PABA → Para-Amino Benzoic Acid.





PABA & Drug are competitors.

Enzyme → rxn specific.

* Fungistatic (Growth stop)
* Fungicidal. (Kill)

Criteria of Antibiotics

① Effective of living system in a very low concentration.

② Very low toxic effect

③ It must be effective against pathogens etc harmful microorganism and noneffective towards non-pathogens

④ Antibiotics should be specific so that storage is easy.

⑤ After having anti biotic it will be excreted fully from body

Mechanism of Action of Antibiotics:

1. By increasing pH or lowering the pH of the cell environment that microorganism cannot tolerate.

2. It can destroy the metabolic system of the pathogens.

3. It impairs the cell division or it can destroy the cell wall of the microorganism.

4. Inhibit cell growth and reproduction.

5. By inhibiting DNA, RNA and protein synthesis.

6. Alternating the surface tension of the cell of the pathogen (Microorganism → Bacteria, fungus)

Classification of Antibiotics based on death and growth of cell:

① Bacteriostatic Antibiotic

These antibiotics can inhibit the growth of the microorganism and hence destroy the bacteria.

Ex— Chloramphenicol, Sulfonamide.

② Bactericidal Antibiotics:

These antibiotics can kill the bacteria through DNA and protein synthesis involves in cell wall construction.

Ex— Penicillin, streptomycin.

Classification of Antibiotics based on number of microorganism

① Broad spectrum Antibiotic:

B.S.A is an antibiotic that acts on the major gram (+)ve or gram (-ve) spe bacteria. Any antibiotic that acts against a wide range of diseases causing by bacteria.

Ex— Tetracycline, chloramphenicol.

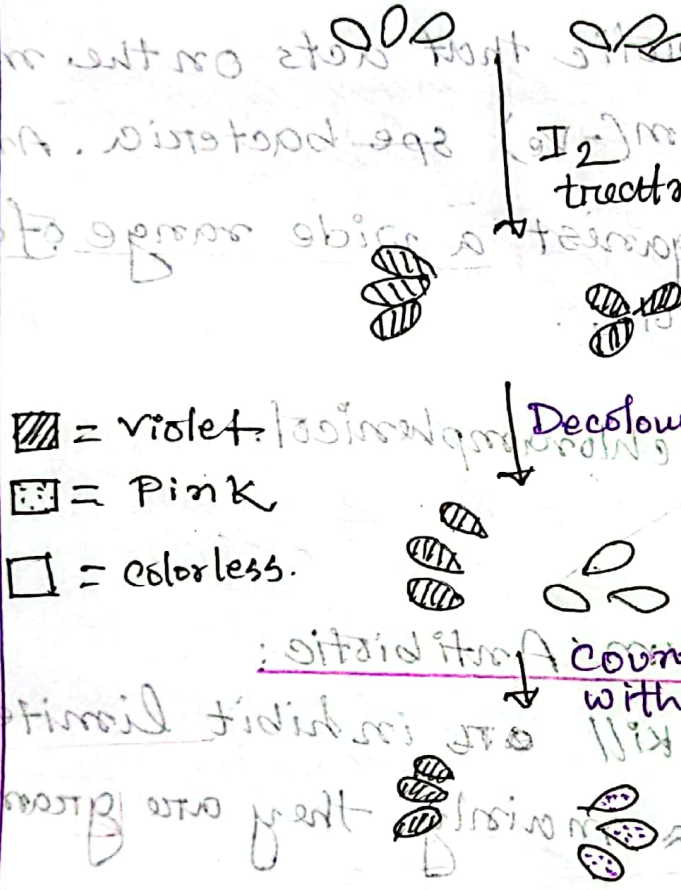
② Narrow spectrum Antibiotic:

N.S.A can able to kill or inhibit limited species of bacteria mainly they are gram (+)ve bacteria.

Ex— Streptomycin, fidaxomycin.

Procedure of Gram - Staining:

Gram (+)ve | Gram (-)ve



The cell wall of Gram (+)ve bacteria is relatively simple structure. It is same 15-50 nm thick and comprises about 50% peptidoglycan, 40-45% acidic polymers, which results in the surface being highly polar and negatively charged together

with 5-10% protein and polysaccharide.. The strongly polar polymers layer influences the penetration of ionized molecule and favours the penetration of cell wall of positively charged compound. These bacteria cell wall have the ability to retain the crystal violet dye after solvent treatment.

Ex - Streptococcus.

The cell wall of Gram (+)ve bacteria is much more complex. It ~~contains~~ consists of a peptido-eglycan layers 2nm in the thickness, forming 5% of the cell wall mass. The outer membrane is a lipid bilayer. similar in some respects to the plasma-membrane, that contain protein molecules. It has complex polysaccharide forming important components of the outer surface. These differ betn stains of bacteria, and there is a periplasmic space containing enzymes and other components.

Ex — Gonococcus.

Penicillin

M.F. — $C_9H_{11}N_2O_4SR$

DBE = 5

Penicillin is a generic term refers a group of organic compound having M.F. $C_9H_{11}N_2O_4SR$; consist with a thiazolidine ring fused with a β-lactam ring.

(1) $C_9H_{11}N_2O_4SR \xrightarrow{NaHCO_3} CO_2 \uparrow$

∴ –COOH present

(2) $C_9H_{11}N_2O_4SR \xrightarrow{NaOH}$ Monosodium salt

Titration করলে 1:1 এ reaction করবে।

(3) $C_9H_{11}N_2O_4SR \xrightarrow{CH_3-\overset{\oplus}{N}\equiv N}$ Mono ester

So, one –COOH is present

$$R-\overset{O}{\overset{\|}{C}}-OH + \ddot{C}H_2-\overset{+}{N}\equiv N \longrightarrow R-\overset{O}{\overset{\|}{C}}-O^{\ominus} + CH_3-\overset{+}{N}\equiv N$$

$$R-\overset{O}{\overset{\|}{C}}-OCH_3 + N_2 \uparrow$$

Mechanism লেখো এই

F-22

④ $C_6H_{11}N_2O_4SR \xrightarrow{FeCl_3}$ No color.

$C_6H_{11}N_2O_4SR \xrightarrow{Na_2[Fe(CN)_5NO]}$ No color.

SH/OH থাকলে violet/Red বা কমলা রঙ হয় কিন্তু SH নেই।

The S is not in the (-SH) form.

⑤ $C_6H_{11}N_2O_4SR \xrightarrow{HNO_2}$ No reaction.

$+NO$ 1°/2°/3° amine

⑥ $C_6H_{11}N_2O_4SR \xrightarrow[\text{Hydrolysis}]{HCl} \underset{\text{Penicillamine}}{C_5H_{11}NO_2S} + C_3H_4NO_2R \xrightarrow{} +CO_2$ Penilo aldehyde

R টি মিলে Acid হলো। কিন্তু, -COOH এরকমের যুক্ত ছিল তার সমস্ত e- with drawing group vicinally যুক্ত আছে। সেজন্য instant reaction হয়ে CO_2 বের হয়।

Penicillamine:

① $C_5H_{11}NO_2S$

② $C_5H_{11}NO_2S \xrightarrow{FeCl_3}$ Colored product.

$\xrightarrow{Na_2[Fe(CN)_5NO]}$

So, SH is present.

NO_2 (sulphonic acid)

$\xrightarrow{Br_2+H_2O} C_5H_{10}$

③ $C_5H_{11}NO_2S \xrightarrow{HNO_2} C_5H_9SO_2(OH) + N_2\uparrow + H_2$

Aliphatic 1° amine.

As alcohol is produced, 1° amine is present.

Aliphatic 1°, 2°, 3°
Aromatic 1°, 2°, 3°
} HNO_2 के साथ Reaction करते हैं।

* Primary aromatic & Aliphatic both gives reaction with HNO_2 and they yields

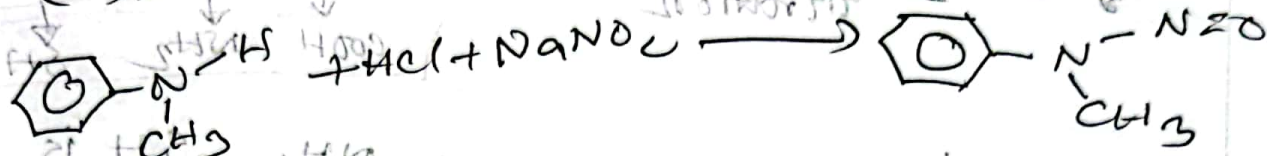
$$Ar-NH_2 + NaNO_2 \xrightarrow[HCl]{cold} Ar-N_2^+X^- + NaX + H_2O$$

$$R-NH_2 + NaNO_2 + HCl \longrightarrow R-N_2^+Cl^- \xrightarrow{H_2O} N_2 + \text{Mixture of alcohol and}$$

[unstable]

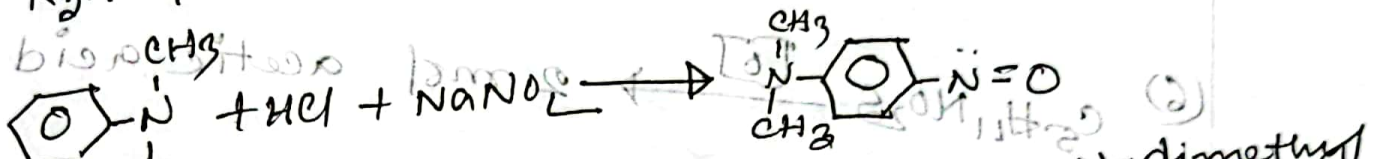
2° amines both aliphatic & aromatic yields N-nitrosoamines —

$(CH_3)_2NH + HCl + NaNO_2 \longrightarrow (CH_3)_2\overset{+}{N}-N=O$



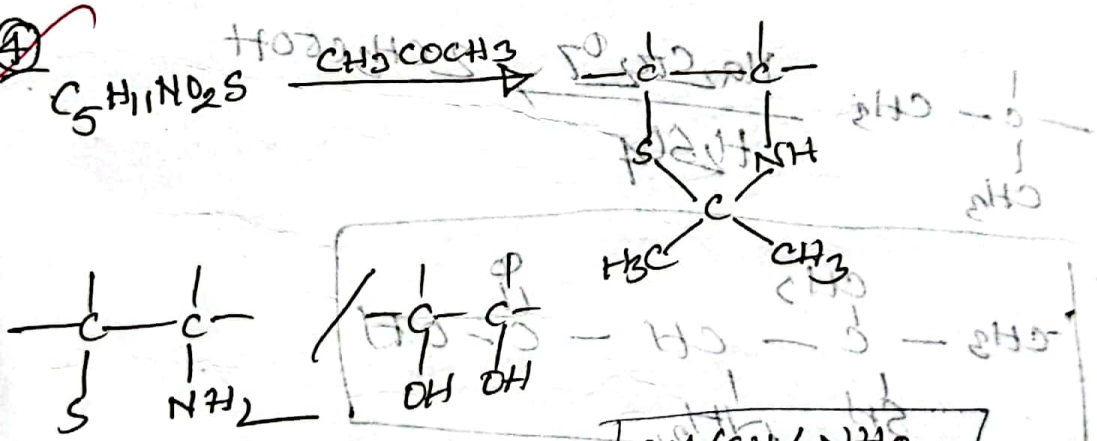
But 3° —

$R_3N + HCl + NaNO_2 \rightleftharpoons R_3\overset{+}{N}HX^- + R_3N-N=OX^-$

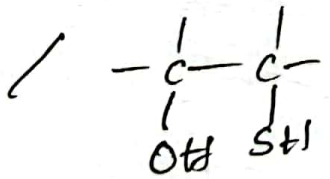


P-Nitroso- N, N-dimethyl aniline

④ $C_5H_{11}NO_2S$ $\xrightarrow{CH_3COCH_3}$



OH/SH/NH_2 vicinal-ला शक्यतो cyclic form होतो.



(5)

$C_5H_{11}NO_2S$ —conductometric titration→ pka = 1.8, 7.9, ~~6.0~~, 10.5

↓ ↓ ↓

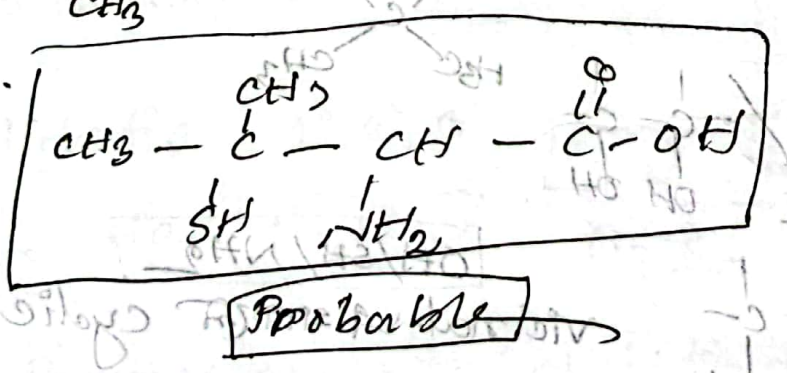
COOH NH_2 SH

So, in penicilline – COOH, NH_2, SH is present

(6) $C_5H_{11}NO_2S$ —[O]→ 2 mol acetic acid

$$-C(=O)-CH_3 \xrightarrow[H_2SO_4]{Na_2Cr_2O_7} CH_3COOH$$

$$-\underset{\underset{CH_3}{|}}{C}-CH_3 \xrightarrow[H_2SO_4]{Na_2Cr_2O_7} 2\,CH_3COOH$$



Probable

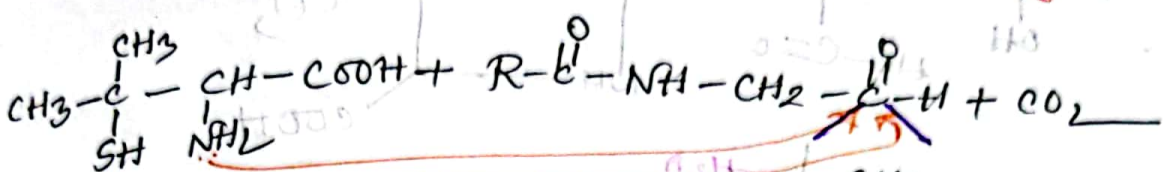
Penicilloaldehyde:

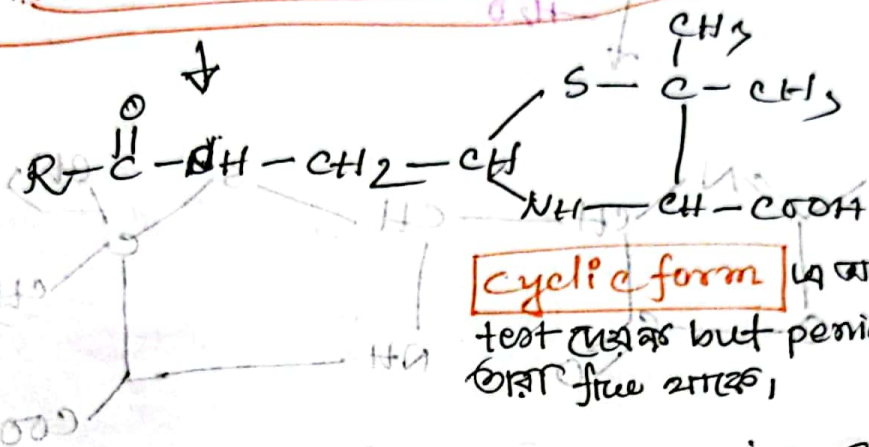
$$C_5H_4NO_2R \xrightarrow[\text{Hydrolysis}]{\text{vigorous}} R\text{-}COOH + NH_2\text{-}CH_2\text{-}C(=O)\text{-}OH$$

So, amide,

$$C_5H_4NO_2R \longrightarrow R\text{-}C(=O)\text{-}NH\text{-}CH_2\text{-}C(=O)\text{-}H$$

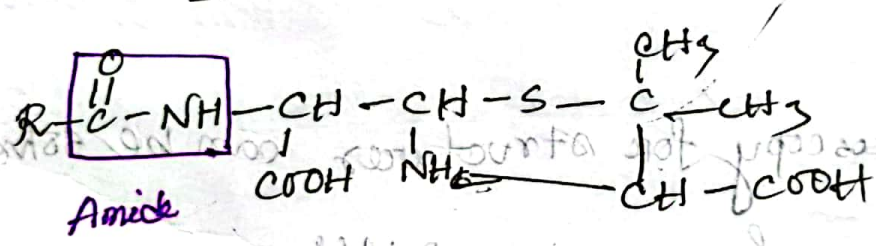
Penicilline → Penicillamine + Penniloaldehyde + CO_2





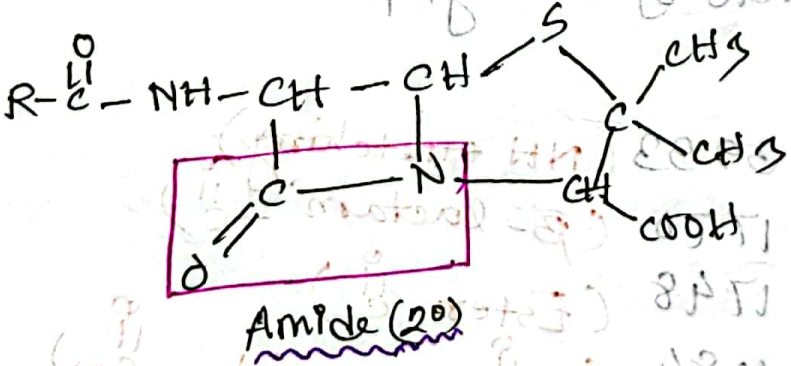
cyclic form এ থাকে তেজস্ক্রিয়া test দেয়ার but penicillamine দেয়া কারণ, তারা free থাকে।

As one CO_2 is evolved; so -COOH is in β position.



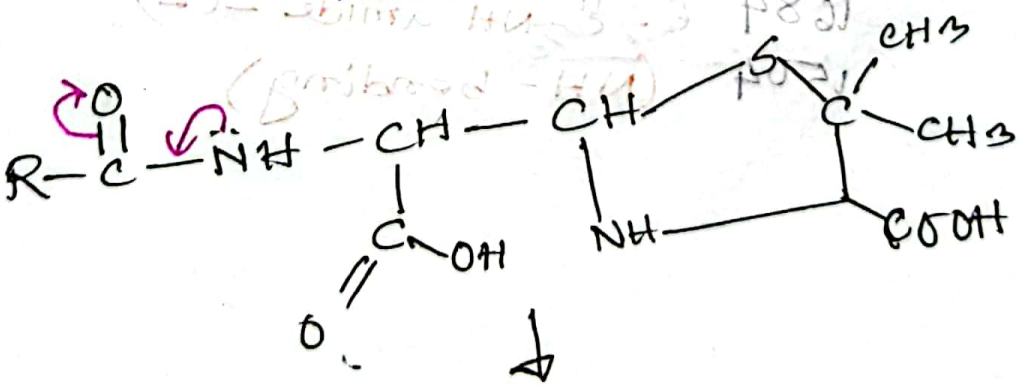
Amide

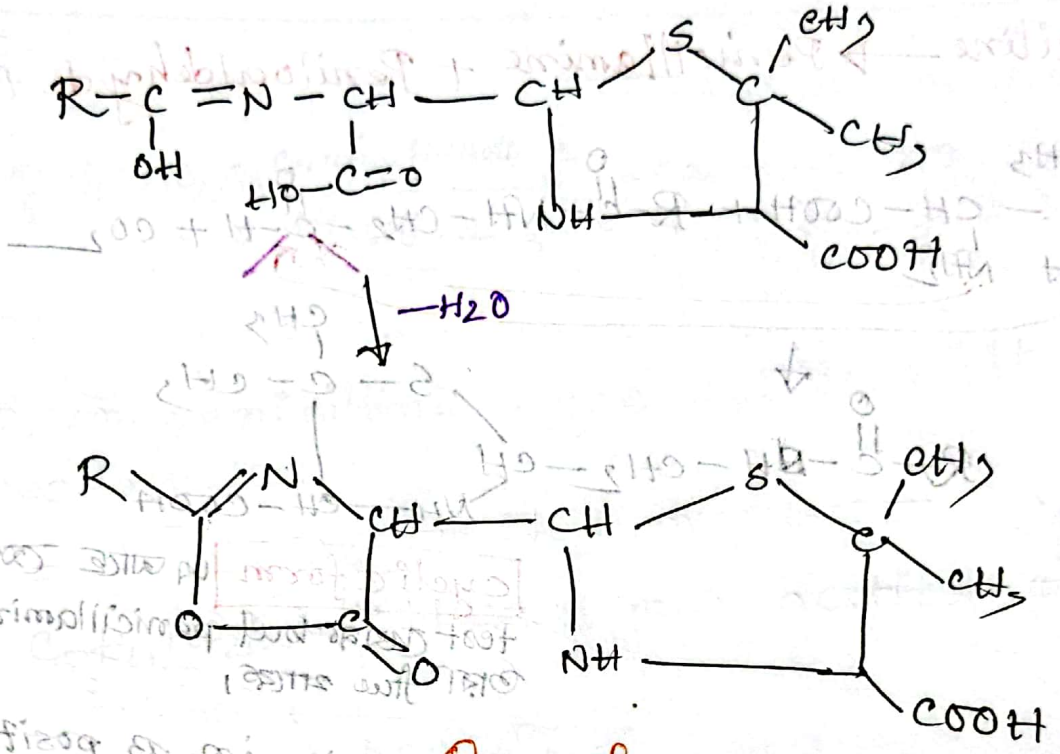
(I) →



Amide (2°)

(II)





Oxazolone

Using spectroscopy for structures can be solved

(1) Methyl ester of benzyl penicillin

IR

3933 (NH stretching)

1770 (β-lactam C=O)

1748 (Ester C=O)

1684 (C=O NH amide C=O)

1504 (NH - bending)

(2) Na-salt of Benzyl penicillin.

IR 3333 (NH salt)

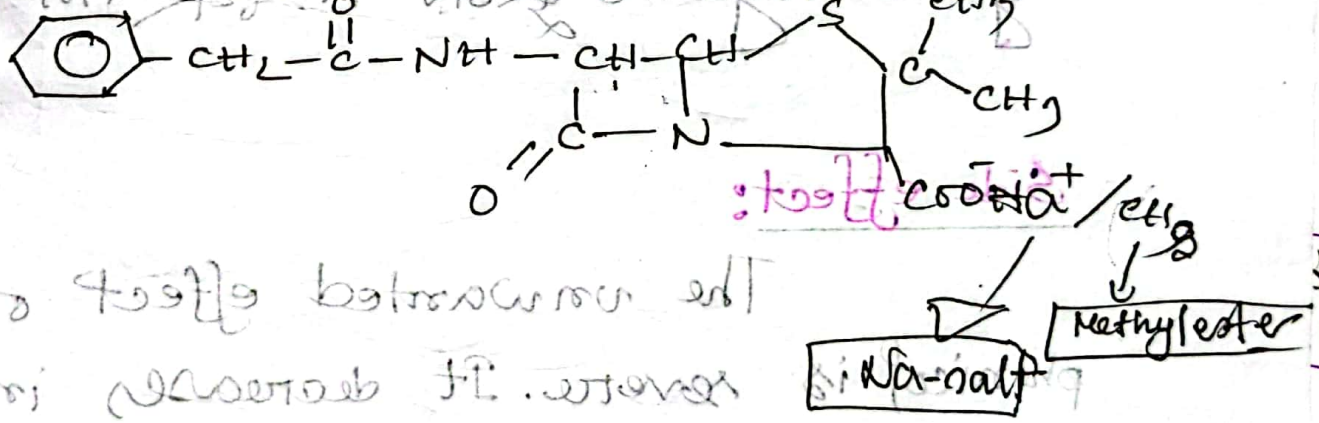
1770 (β-lactam -C=O)

1613 (COO^-, C=O)

1515 (N-H bending)

C=N → 1675 Absent

C=O → 1825 "

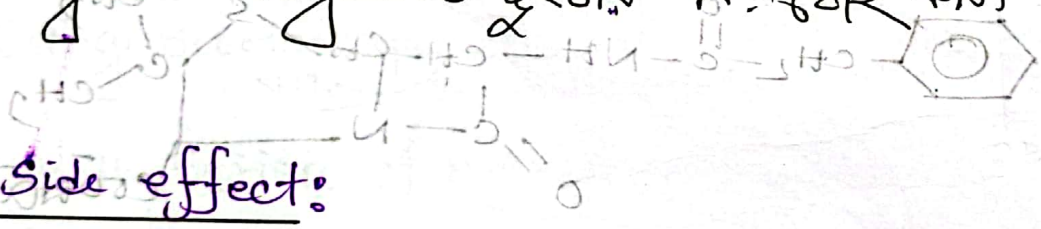


CHLORAMPHENICOL

Broad spectrum antibiotic.
Bacteriostatic for most organism but kills bacteria too by producing chloramphenical acyltransferase.

Q. যেমন structure determine করে যে— bacteria kill করে growth rate stop করে

* Broad spectrum — gram positive and gram negative দুইটার বিরুদ্ধেই কাজ করতে পারে।



Side effect:

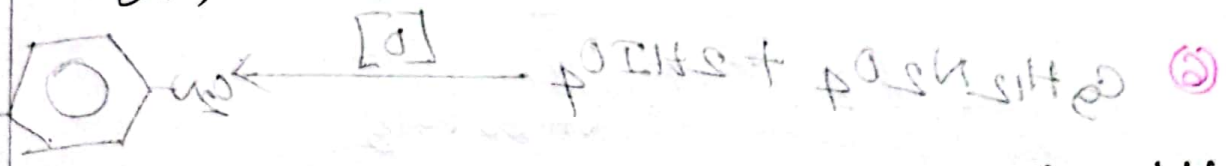
The unwanted effect of chloramphenicol is severe. It decreases in all blood cell element.

$C_{11}H_{12}Cl_2N_2O_5$

DBE v = 6

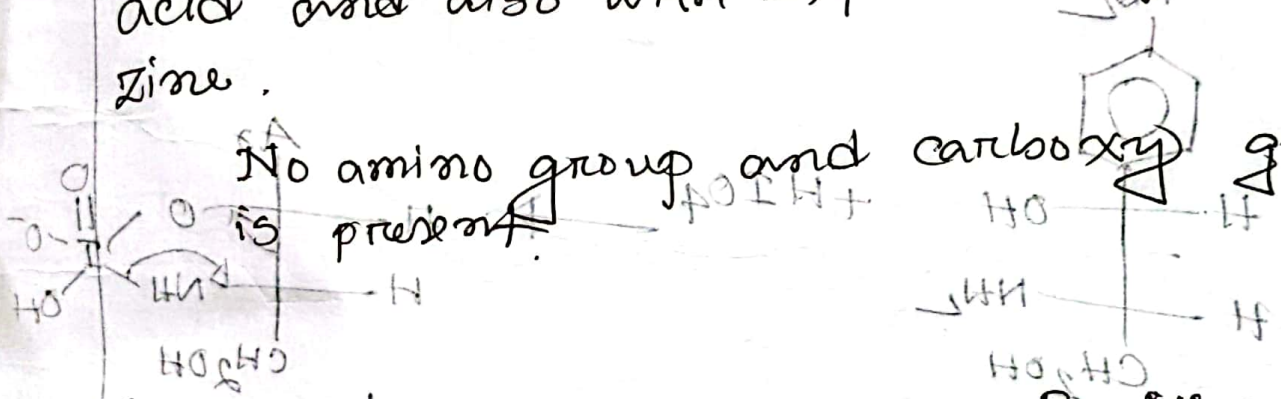
1. $C_{11}H_{12}Cl_2N_2O_5$ reacts with Sn/HCl get reduced and the product reacts with β-naphthol to produce red ppt.

So, NO_2 is present.



2. $C_{11}H_{12}Cl_2N_2O_5$ does not react with HNO_2 acid and also with 2,4-dinitrophenyl hydrazine.

No amino group and carboxyl group is present.

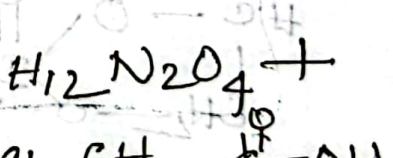


3. $C_{11}H_{12}Cl_2N_2O_5 + (CH_3CO)_2O \xrightarrow{\text{Pyridine}}$ Diacetylated product.

two alcoholic –OH present.



4. $C_{11}H_{12}Cl_2N_2O_5 + H_2O \longrightarrow C_9H_{12}N_2O_4 +$

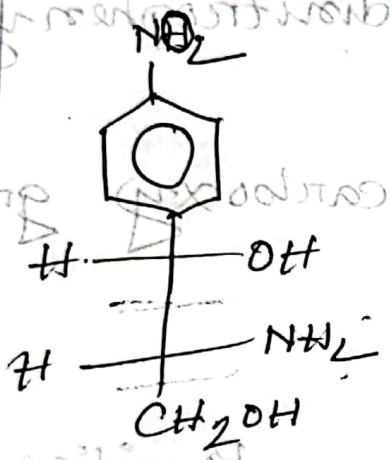


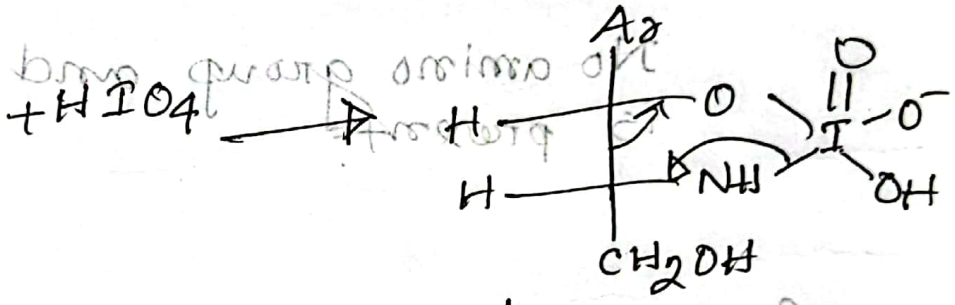
Optically active base.

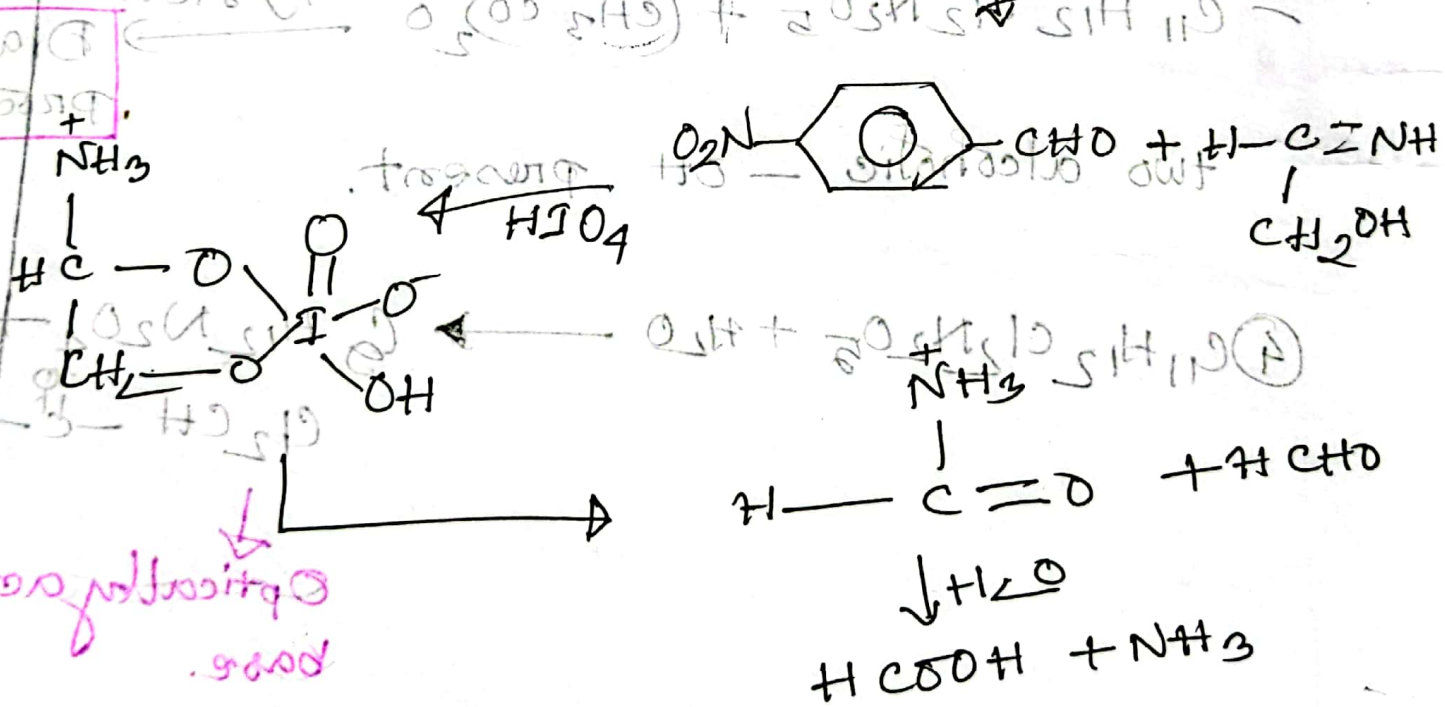
⑤ $C_9H_{12}N_2O_4 + FeCl_3 \longrightarrow$ No reaction.

No phenolic –OH is present.

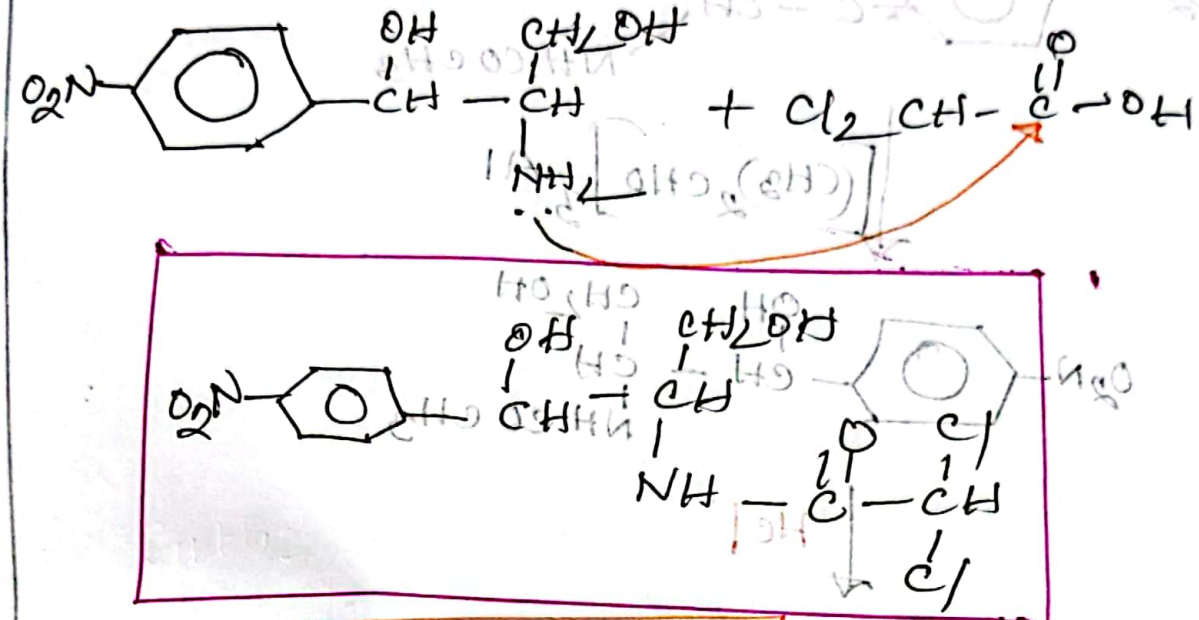
⑥ $C_9H_{12}N_2O_4 + 2HIO_4 \xrightarrow{[O]} O_2N\text{-}C_6H_4\text{-}CHO + HCHO + HCOOH + NH_3$





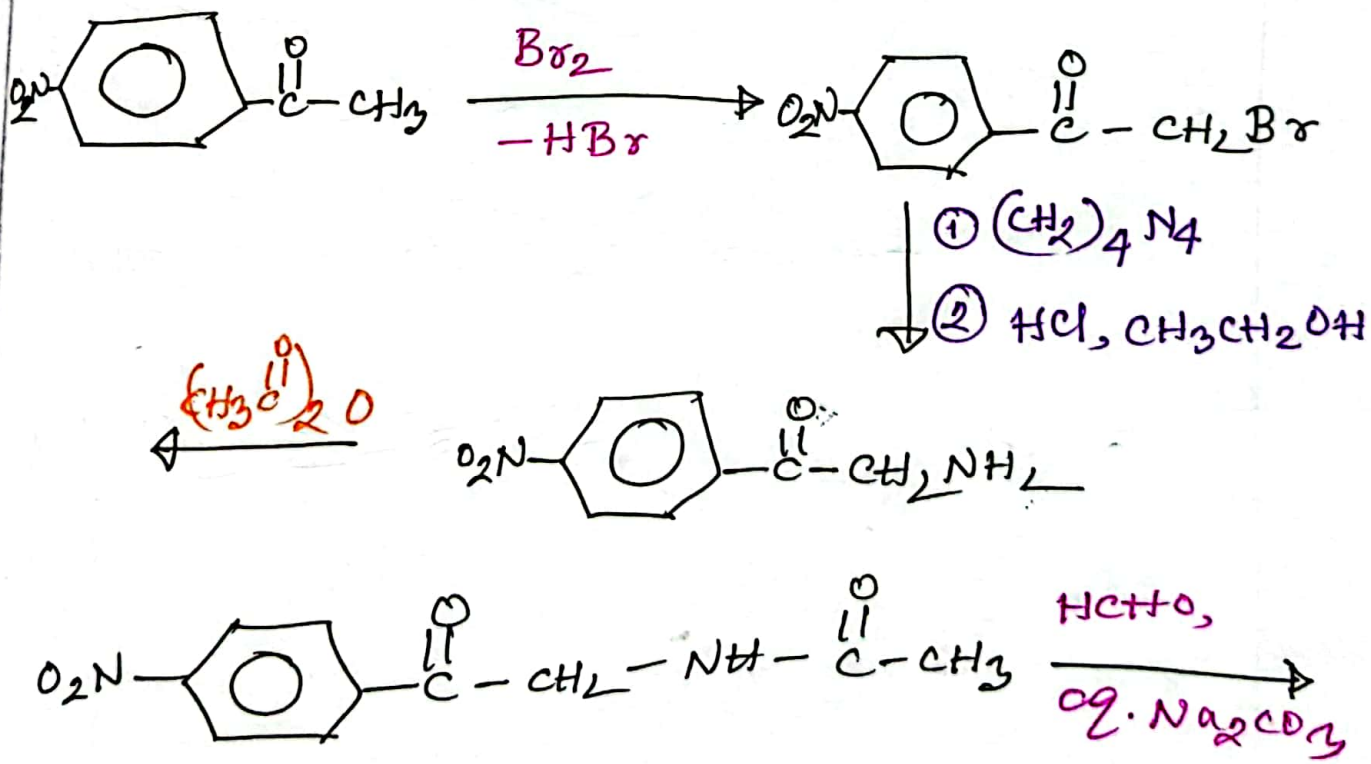


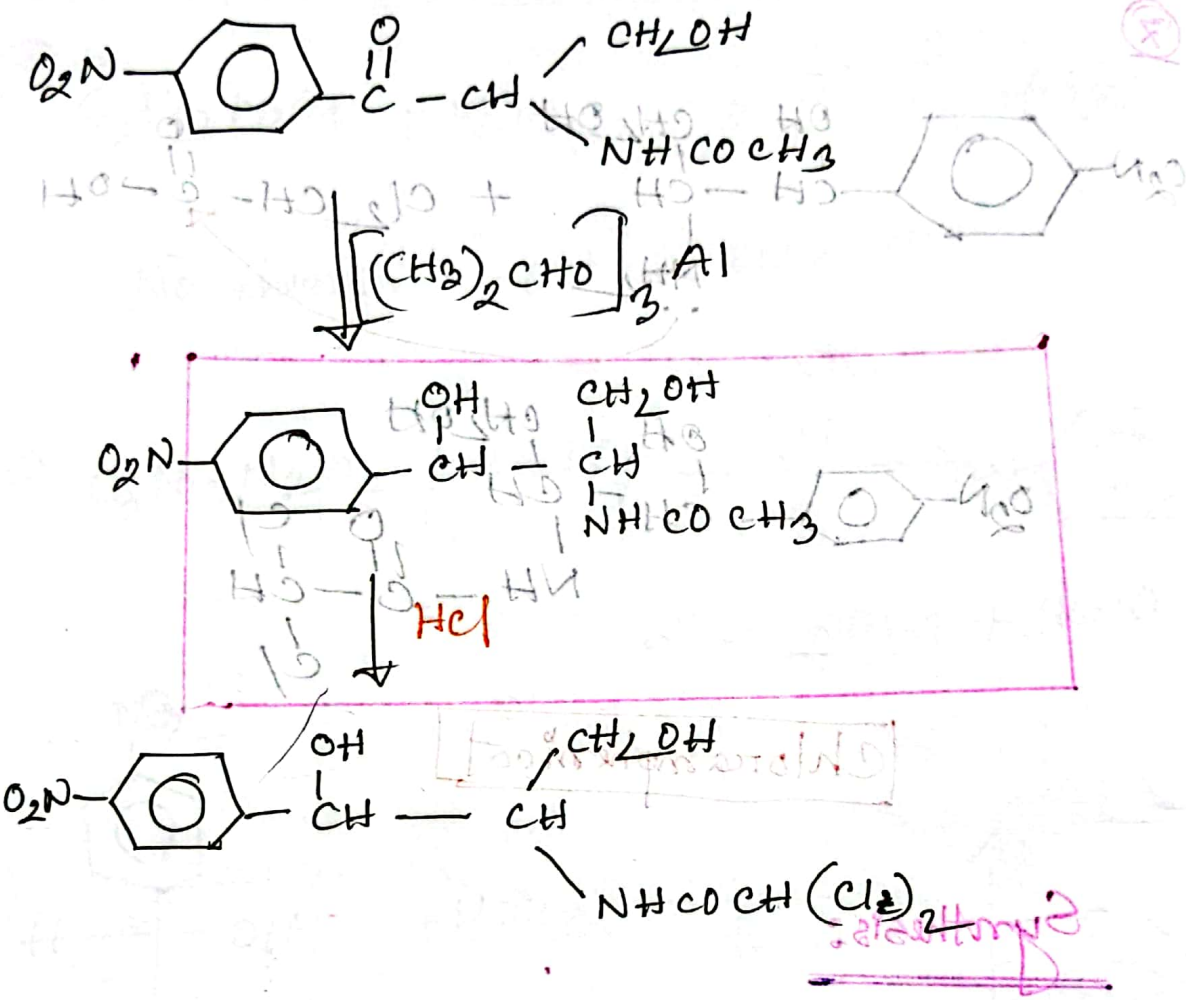
(7)



Chloramphenicol

Synthesis:




O_2N
C
O
CH
CH_2OH
$NHCOCH_3$
$[(CH_3)_2CHO]_3Al$
OH
CH_2OH
HCl
$NHCOCH(Cl_2)$